

COMPANION DIGITAL PACKAGE

Identity Lab Setup Guide

Build your own identity lab — hands-on practice for CIDPro™

8 step-by-step lab guides covering Keycloak, OpenLDAP, Okta, Entra ID, HashiCorp Vault and more Free and open-source tools you can run on your own machine

Mastering Identity Security for CIDPro™
Abbas Kudrati | 2026 Edition

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Build Your Own Identity Lab

A Practical Setup Guide for Practising Identity Tools at Home Using Free and Commercial Tools

CIDPro™ Exam Prep Book

Why Hands-On Practice Is Essential for Identity Professionals

Reading about identity concepts is important, but it is not sufficient. Identity and access management is an applied discipline—the difference between understanding OAuth 2.0 in theory and correctly configuring an OIDC relying party is the difference between passing an exam and excelling in a real-world role. Every misconfigured conditional access policy, every expired certificate that causes an outage, and every poorly designed RBAC model teaches lessons that no textbook can.

Building a personal identity lab gives you a safe environment to experiment, break things, and learn from mistakes without risking production systems. It allows you to explore tools referenced throughout this book—from Keycloak and OpenLDAP to Microsoft Entra ID and CyberArk—and to validate your understanding of the CIDPro objectives through direct experience.

The good news: you do not need expensive equipment or enterprise licences to get started. Nearly every major identity platform offers free tiers, developer editions, or trial accounts specifically designed for learning.

Hardware and Cloud Requirements

You can run a fully functional identity lab on modest hardware. A modern laptop or desktop with at least 8 GB of RAM (16 GB recommended), 50 GB of free disk space, and a multi-core processor is sufficient for running virtual machines and containers locally. For cloud-based labs, all three major cloud providers (AWS, Azure, Google Cloud) offer free tiers that include enough resources to practise identity configurations at no cost.

Recommended local setup: Install Docker Desktop (free for personal use) or a hypervisor such as VirtualBox (free) or VMware Workstation Player (free for non-commercial use). Docker is the fastest path to running Keycloak, OpenLDAP, HashiCorp Vault, and other identity tools with minimal overhead.

Recommended cloud setup: Create free-tier accounts on Microsoft Azure (includes Entra ID), AWS, and Google Cloud. Combine these with Okta Developer Edition and Auth0's free tier to cover the full spectrum of workforce and customer identity scenarios.

Docker Desktop: <https://www.docker.com/products/docker-desktop/>

VirtualBox: <https://www.virtualbox.org/>

Section D1: Free and Open-Source Identity Tools to Practise With

The open-source identity ecosystem is remarkably rich. The tools below cover the full range of identity capabilities—from authentication and federation to directory services, IGA, PKI, and secrets management. Each can be installed locally using Docker or virtual machines.

Table D.1 — Free and Open-Source Identity Tools

Tool	What It Does	Best For Practising	URL
Keycloak	Open-source identity provider with SSO, OIDC, SAML, and user federation	Authentication, federation, OIDC/SAML configuration, identity brokering, social login	keycloak.org
Gluu Server	Enterprise-grade open-source IAM with FIDO2 support	Full IAM stack, FIDO2 passkey testing, SCIM provisioning	gluu.org
OpenLDAP	Open-source directory service implementing LDAP protocol	Directory services, LDAP queries, schema design, user management	openldap.org
FreeIPA	Linux-based identity management with Kerberos, LDAP, DNS, and certificate management	Linux IAM, Kerberos authentication, policy management, certificate services	freeipa.org
HashiCorp Vault	Secrets management, encryption as a service, and PKI engine	Machine identity, NHI secrets, PKI certificate issuance, dynamic credentials	vaultproject.io
OpenIAM	Open-source identity governance and administration platform	IGA, access reviews, provisioning workflows, role management	openiam.com
EJBCA	Open-source enterprise PKI and certificate authority	PKI, certificate lifecycle, X.509 certificates, CRL/OCSP	ejbca.org
Authentik	Modern open-source IdP with flows, OIDC, SAML, LDAP proxy	Custom authentication flows, self-service, OIDC/SAML, SCIM	goauthentik.io
Ory (Kratos, Hydra)	Cloud-native identity infrastructure: user mgmt (Kratos), OAuth2 (Hydra)	Developer-focused identity, OAuth2/OIDC server, API-first identity	ory.sh
Apache Syncope	Open-source identity management with provisioning and workflow	Identity provisioning, connectors, workflow-based approvals	syncope.apache.org

Lab Tip: Keycloak is the single most versatile tool for CIDPro exam preparation. It lets you practise SSO, federation, OIDC, SAML, user federation with LDAP, social login, and custom authentication flows—all from a single Docker container.

Section D2: Free Tiers and Trial Accounts for Commercial Tools

Every major commercial identity vendor offers some form of free access for learning and development. These free tiers let you practise with the same enterprise tools used in production environments worldwide—giving you hands-on experience that directly translates to job-ready skills.

Table D.2 — Free Tiers and Trial Accounts

Tool	Free Access Option	What to Practise	URL
Okta	Developer Edition (free forever)	SSO, OIDC, MFA, SCIM provisioning, API access management, lifecycle hooks	developer.okta.com/signup
Microsoft Entra ID	M365 Developer Program (free, renewable 90-day)	Entra ID, Conditional Access, PIM, app registrations, B2B collaboration, identity governance	developer.microsoft.com/microsoft-365/dev-program
Ping Identity	PingOne trial (30-day)	SSO, MFA, CIAM, DaVinci orchestration, risk-based authentication	pingidentity.com/en/try-ping
CyberArk	Privilege Cloud trial (30-day)	PAM, credential vaulting, session recording, just-in-time access	cyberark.com/free-trial
SailPoint	Identity University (free courses + sandbox)	IGA, access reviews, certification campaigns, role modelling, provisioning	university.sailpoint.com
Auth0 (by Okta)	Free tier (up to 25,000 MAU)	CIAM, OIDC, social login, passwordless, Actions extensibility, custom domains	auth0.com/signup
AWS IAM	Free tier (always free for IAM)	Cloud IAM policies, roles, identity federation, IAM Access Analyzer, Organizations	aws.amazon.com/free
Google Cloud IAM	Free tier (\$300 credit for 90 days)	Cloud IAM, service accounts, workload identity, organisation policies	cloud.google.com/free
Azure AD B2C	Free up to 50,000 MAU	CIAM, social identity providers, custom user flows, API connectors, branding	azure.microsoft.com/services/active-directory-b2c
Descope	Free tier (up to 7,500 MAU)	Passwordless, passkeys, CIAM, drag-and-drop authentication flows	descope.com

Cost Tip: You can build a comprehensive multi-vendor identity lab for zero cost by combining Okta Developer Edition + Microsoft 365 Developer Program + Auth0 free tier + AWS free tier + a local Docker instance of Keycloak. This covers workforce SSO, CIAM, cloud IAM, and open-source identity.

Section D3: Step-by-Step Home Lab Setup Guides

The following eight labs walk you through practical identity scenarios. Each lab is designed to be completed in one to three hours and maps directly to CIDPro exam objectives. Detailed step counts are kept concise—follow the linked references for deeper walkthroughs.

Lab 1: Set Up Keycloak and Configure SSO with SAML

CIDPro Objective: 2.2 (Authentication), 4.3 (IAM Standards — SAML)

Time: 60–90 minutes

Step 1: Install Keycloak. Run: `docker run -p 8080:8080 -e KC_BOOTSTRAP_ADMIN_USERNAME=admin -e KC_BOOTSTRAP_ADMIN_PASSWORD=admin quay.io/keycloak/keycloak:latest start-dev`

Step 2: Create a Realm. Log in at `http://localhost:8080`, create a new realm (e.g., "identity-lab"), and add test users with passwords.

Step 3: Register a SAML Client. Add a new client using the SAML protocol. Set the client ID, valid redirect URIs, and download the SAML metadata XML.

Step 4: Configure a Sample App. Deploy a simple SAML SP (e.g., using the Spring Security SAML sample or a Python flask-saml2 app) and point it at your Keycloak realm's SAML metadata endpoint.

Step 5: Test SSO. Log in via the sample app, observe the SAML assertion in browser dev tools, and verify user attributes are passed correctly.

Keycloak Getting Started: <https://www.keycloak.org/getting-started/getting-started-docker>

Keycloak SAML Guide: https://www.keycloak.org/docs/latest/server_admin/#saml-clients

Lab 2: Connect Keycloak to OpenLDAP as a User Directory

CIDPro Objective: 2.1 (Identity Lifecycle), 4.3 (LDAP)

Time: 60–90 minutes

Step 1: Run OpenLDAP. Use the osixia/openldap Docker image: `docker run -p 389:389 -e LDAP_ORGANISATION="Identity Lab" -e LDAP_DOMAIN="lab.local" -e LDAP_ADMIN_PASSWORD=admin osixia/openldap:latest`

Step 2: Populate Directory. Use Apache Directory Studio (free GUI) or `ldapadd` to create organisational units and user entries under `dc=lab,dc=local`.

Step 3: Configure User Federation. In Keycloak Admin Console, go to User Federation, add an LDAP provider, and configure the connection URL (`ldap://host.docker.internal:389`), bind DN, and user search base.

Step 4: Sync and Test. Trigger a full sync, then verify that LDAP users appear in Keycloak and can authenticate through the Keycloak login page.

Step 5: Explore Mappers. Configure attribute mappers to control which LDAP attributes are exposed as Keycloak user attributes and token claims.

Apache Directory Studio: <https://directory.apache.org/studio/>

Keycloak LDAP Federation: https://www.keycloak.org/docs/latest/server_admin/#_ldap

Lab 3: Set Up Okta Developer Edition and Build an OIDC Application

CIDPro Objective: 2.2 (Authentication), 4.3 (OIDC, OAuth 2.0)

Time: 45–60 minutes

Step 1: Sign Up. Create a free Okta Developer Edition account at developer.okta.com/signup.

Step 2: Create an Application. In the Okta Admin Console, navigate to Applications > Create App Integration. Select OIDC as the sign-in method and Web Application as the type. Note the client ID and secret.

Step 3: Build a Sample App. Use Okta's quickstart guides to create a simple web application in your preferred language (Node.js, Python, Java, or .NET) that uses the Authorization Code flow with PKCE.

Step 4: Configure MFA. Enable a second factor (Okta Verify or SMS) in Security > Authenticators, and add an authentication policy that requires MFA for your application.

Step 5: Inspect Tokens. Use jwt.io or Okta's token preview to decode the ID token and access token. Identify standard OIDC claims such as sub, iss, aud, and exp.

Okta Developer Portal: <https://developer.okta.com/>

Okta OIDC Quickstarts: <https://developer.okta.com/docs/guides/>

Lab 4: Configure Microsoft Entra ID Conditional Access and MFA

CIDPro Objective: 2.2 (Authentication), 3.5 (Operational Considerations)

Time: 60–90 minutes

Step 1: Join the Developer Program. Sign up at developer.microsoft.com/microsoft-365/dev-program to get a free Microsoft 365 E5 sandbox with Entra ID P2 features.

Step 2: Create Test Users. In the Entra admin centre (entra.microsoft.com), create several test users and assign them to groups (e.g., "Finance", "Engineering").

Step 3: Configure Conditional Access. Create a Conditional Access policy that requires MFA for the "Finance" group when accessing SharePoint. Set a second policy that blocks access from a specific named location.

Step 4: Enable PIM. Activate Privileged Identity Management for the Global Administrator role. Configure it to require approval and justification for elevation.

Step 5: Test and Review Logs. Sign in as a test user, trigger the Conditional Access policies, and review the sign-in logs and audit logs in Entra to understand how policies are evaluated.

Microsoft 365 Developer Program: <https://developer.microsoft.com/en-us/microsoft-365/dev-program>

Conditional Access Documentation: <https://learn.microsoft.com/en-us/entra/identity/conditional-access/>

Lab 5: Deploy HashiCorp Vault for Secrets Management and PKI

CIDPro Objective: 2.3 (Access Control), 3.6 (Threat Modelling — NHI)

Time: 60–90 minutes

Step 1: Install Vault. Run in dev mode: `docker run --cap-add=IPC_LOCK -p 8200:8200 hashicorp/vault:latest server -dev -dev-root-token-id=root`

Step 2: Store and Retrieve Secrets. Using the Vault CLI or UI at <http://localhost:8200>, enable the KV secrets engine and store a sample API key. Retrieve it using the CLI and the HTTP API.

Step 3: Enable the PKI Engine. Mount the PKI secrets engine, generate a root CA, and configure a role for issuing certificates.

Step 4: Issue a Certificate. Request a certificate for a test hostname. Examine the certificate, private key, and CA chain returned by Vault.

Step 5: Explore Dynamic Secrets. Enable the database secrets engine with a test PostgreSQL instance and generate dynamic, short-lived database credentials.

Vault Getting Started: <https://developer.hashicorp.com/vault/tutorials/getting-started>

Vault PKI Tutorial: <https://developer.hashicorp.com/vault/tutorials/secrets-management/pki-engine>

Lab 6: Simulate a PAM Workflow Using CyberArk Privilege Cloud Trial

CIDPro Objective: 5.2 (Privilege Management Operations)

Time: 90–120 minutes

Step 1: Request a Trial. Sign up for a CyberArk Privilege Cloud trial at cyberark.com. You will receive a dedicated tenant with full PAM capabilities.

Step 2: Onboard a Privileged Account. Add a test Windows or Linux privileged account to the vault. Configure the platform, safe, and access permissions.

Step 3: Configure Just-in-Time Access. Set up a workflow that requires a request and approval before a privileged credential can be checked out.

Step 4: Connect and Record. Use the CyberArk PSM (Privileged Session Manager) to initiate a session. Observe how the session is recorded and auditable.

Step 5: Review Audit Logs. Examine the audit trail in the CyberArk console to see who accessed what, when, and what actions were taken during the session.

CyberArk Free Trial: <https://www.cyberark.com/try/>

CyberArk Documentation: <https://docs.cyberark.com/>

Lab 7: Build an Access Review Workflow in SailPoint Identity University

CIDPro Objective: 2.4 (Identity Governance), 5.2 (Privilege Management)

Time: 60–90 minutes

Step 1: Enrol. Create a free account at university.sailpoint.com and navigate to the hands-on lab environment.

Step 2: Explore the Sandbox. Familiarise yourself with the SailPoint IdentityNow interface: identities, access profiles, roles, and entitlements.

Step 3: Create a Certification Campaign. Define an access certification campaign that reviews all entitlements for a specific department. Set reviewers, deadlines, and escalation rules.

Step 4: Execute the Review. Act as a reviewer: approve, revoke, or reassign access for individual users. Observe how decisions trigger downstream provisioning actions.

Step 5: Analyse Results. Review the campaign dashboard for compliance metrics, completion rates, and remediation actions taken.

SailPoint University: <https://university.sailpoint.com/>

SailPoint Documentation: <https://documentation.sailpoint.com/>

Lab 8: Set Up Auth0 for CIAM with Social Login

CIDPro Objective: 2.2 (Authentication), 3.3 (CIAM)

Time: 45–60 minutes

Step 1: Sign Up. Create a free Auth0 account at auth0.com/signup. You get a tenant with up to 25,000 monthly active users at no cost.

Step 2: Enable Social Connections. In Authentication > Social, enable Google and GitHub social login providers. You will need to register OAuth apps with each provider (Auth0 provides dev keys for testing).

Step 3: Create a Sample Application. Register a Single Page Application or Regular Web App in Auth0 and use the Auth0 quickstart to scaffold a sample app in React, Next.js, or Python.

Step 4: Customise the Login Flow. Use Auth0 Actions (post-login) to add custom logic—for example, enriching user profiles with metadata or blocking logins from specific email domains.

Step 5: Test Progressive Profiling. Configure step-up authentication that prompts users for additional information (e.g., phone number) on their second login, simulating progressive profiling in a CIAM scenario.

Auth0 Documentation: <https://auth0.com/docs>

Auth0 Quickstarts: <https://auth0.com/docs/quickstarts>

Section D4: Recommended Learning Platforms to Pair with Your Lab

Hands-on labs are most effective when paired with structured learning content. The following platforms offer free and paid courses that align directly with identity lab exercises and CIDPro exam preparation.

Table D.3 — Recommended Learning Platforms

Platform	What It Covers	Cost	URL
Microsoft Learn	Entra ID, SC-300 exam prep, Conditional Access, PIM, identity governance modules	Free	learn.microsoft.com
SailPoint University	IGA fundamentals, IdentityNow administration, certification exam prep	Free	university.sailpoint.com
CyberArk Education	PAM fundamentals, Trustee-level training, Privilege Cloud basics	Free (foundation)	training.cyberark.com
Okta Learning Portal	Okta basics, OIDC/SAML configuration, developer guides, certification prep	Free (Dev Edition users)	developer.okta.com
HashiCorp Learn	Vault tutorials, PKI, dynamic secrets, secrets management best practices	Free	developer.hashicorp.com/vault/tutorials
AWS Skill Builder	Cloud IAM, security specialty prep, hands-on labs	Free + paid	skillbuilder.aws
Google Cloud Skills Boost	GCP IAM, service accounts, Workload Identity, security engineer prep	Free + paid	cloudskillsboost.google
IDPro Body of Knowledge	Vendor-neutral IAM concepts, the authoritative reference for CIDPro exam preparation	Free (open access)	bok.idpro.org
Pluralsight	Broad IAM coverage, identity architecture, Zero Trust, cloud security paths	Paid subscription	pluralsight.com
Udemy / LinkedIn Learning	On-demand IAM courses, vendor certification prep, practical tutorials	Paid (per course / sub)	udemy.com linkedin.com/learning

Pro Tip: Start with the free platforms (Microsoft Learn, SailPoint University, IDPro BoK) to build your conceptual foundation, then use your home lab to apply what you have learned. Theory plus practice is the formula that turns knowledge into competence.

Your identity lab is a living resource that grows with your career. As you progress from early-career practitioner to architect and leader, revisit and expand your lab to include new tools, new protocols, and new challenges. The professionals who invest in continuous hands-on practice are the ones who lead the identity profession forward.